Givens:

Alert or event score $\begin{cases} \text{if score} \geq 0{,}6 \text{ (flaged)} \\ \text{else ignored} \end{cases}$

Threshold $= 0{,}6$ } 300 alertes per day

A $\begin{cases} x = 0{,}6 \\ y = 90 \end{cases}$

Treshold $+ 0{,}1 \Rightarrow -15$

solution:

1- The decision variable here is x (the threshold).

2) The function of number of attack detected denote

$D(x) = ax + b \Rightarrow a(x - 0{,}6) + b$ ($b = 90$)

$a = \frac{\Delta y}{\Delta x} = \frac{-15}{+0{,}1} = -150$, $D(x) = 90 - 150(x - 0{,}6)$

$A(x) = ax + b \Rightarrow a(x - 0{,}6) + b$

$b = 300$

$a = \frac{\Delta y}{\Delta x} = \frac{-200}{+0{,}1} \Rightarrow -2000$

$A(x) = -2000(x - 0{,}6) + 300$

$\begin{cases} A(x) = -2000x + 1500 \\ D(x) = -150x + 180 \end{cases}$

$A(x) = $ Alert

$D(x) = $ Detected

3)

3) The benefit function

$g(x) = 500\,D(x) - 100\,A(x)$

$g(x) = 500(-150x + 180) - 100(-2000x + 1500)$

$g(x) = 125\,000x - 60\,000$

4) Neened Constraints

$0 \leq x \leq 1$

5) solve the optimization to maintain 80 deteted attack per day

$A(x) \geq 80 \Rightarrow -2000x + 1500 \geq 80$

$\Rightarrow x \leq \frac{1500 - 80}{2000}$

$\Rightarrow x \leq 0{,}71$

$\Rightarrow x \in [0{,}6 ; 0{,}71]$

The detected attack number:

$D(x) = -150x + 180$, let's $x = 0{,}71$

$D(0{,}71) = -150(0{,}71) + 180$

≈ 74

Interpretation:

The volume of alerts reduce from 200 to 74 per day.